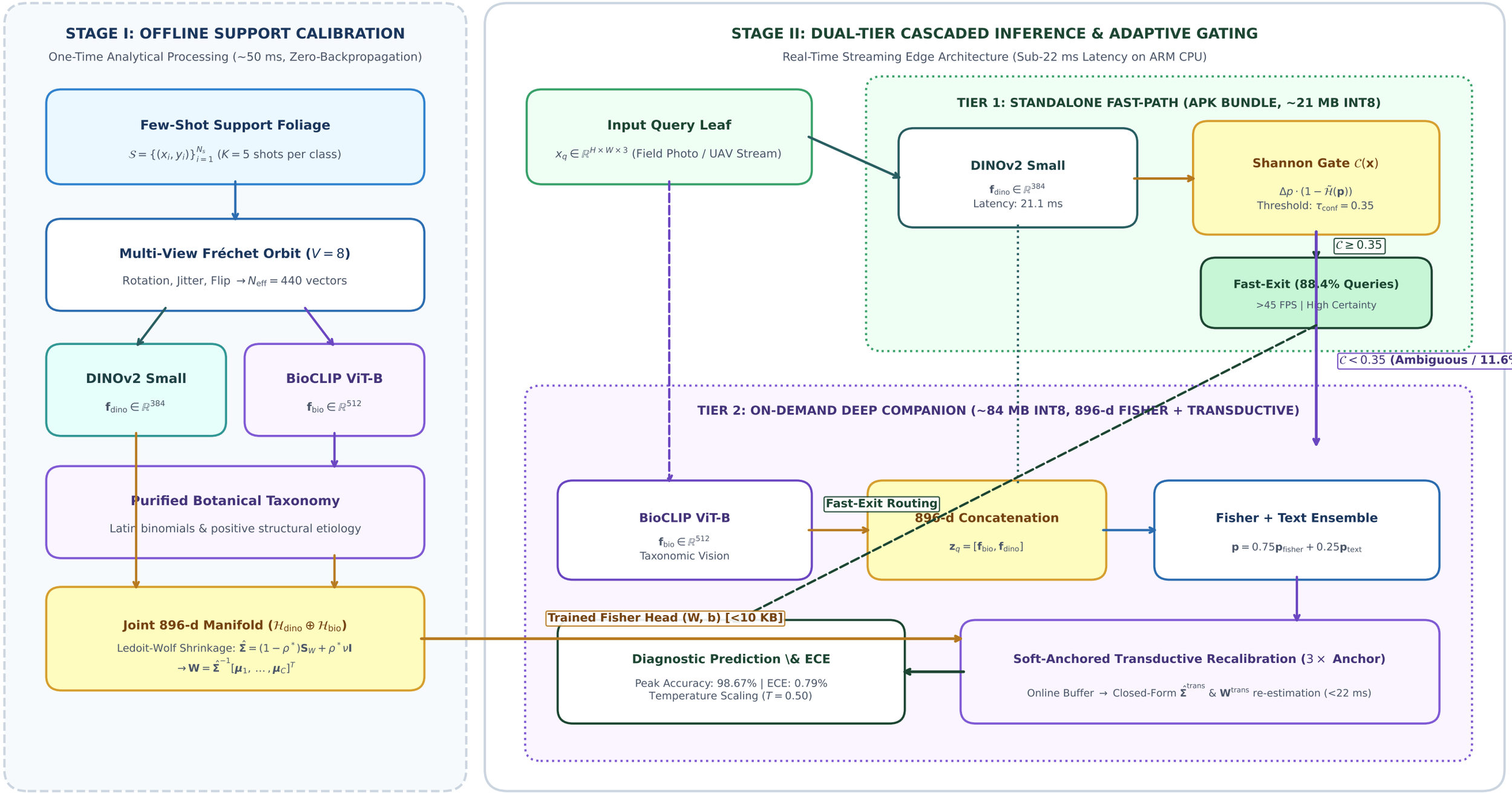

STAGE I: OFFLINE SUPPORT CALIBRATION
One-Time Analytical Processing (~50 ms, Zero-Backpropagation)
Few-Shot Support Foliage
$\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{N_s}$ ($K = 5$ shots per class)
Multi-View Fréchet Orbit ($V = 8$)
Rotation, Jitter, Flip $\rightarrow N_{\text{eff}} = 440$ vectors
DINOv2 Small
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$
BioCLIP ViT-B
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$
Purified Botanical Taxonomy
Latin binomials & positive structural etiology
Joint 896-d Manifold ($\mathcal{H}_{\text{dino}} \oplus \mathcal{H}_{\text{bio}}$)
Ledoit-Wolf Shrinkage: $\hat{\mathbf{\Sigma}} = (1 - \rho^*)\mathbf{S}_W + \rho^* \nu \mathbf{I}$
$\rightarrow \mathbf{W} = \hat{\mathbf{\Sigma}}^{-1}[\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_C]^T$
STAGE II: DUAL-TIER CASCADED INFERENCE & ADAPTIVE GATING
Real-Time Streaming Edge Architecture (Sub-22 ms Latency on ARM CPU)
Input Query Leaf
$x_q \in \mathbb{R}^{H \times W \times 3}$ (Field Photo / UAV Stream)
TIER 1: STANDALONE FAST-PATH (APK BUNDLE, ~21 MB INT8)
DINOv2 Small
$\mathbf{f}_{\text{dino}} \in \mathbb{R}^{384}$
Latency: 21.1 ms
Shannon Gate $\mathcal{C}(\mathbf{x})$
$\Delta p \cdot (1 - \tilde{\mathcal{H}}(\mathbf{p}))$
Threshold: $\tau_{\text{conf}} = 0.35$
$\mathcal{C} \geq 0.35$
Fast-Exit (88.4% Queries)
>45 FPS | High Certainty
$\mathcal{C} < 0.35$ (Ambiguous / 11.6%)
TIER 2: ON-DEMAND DEEP COMPANION (~84 MB INT8, 896-d FISHER + TRANSDUCTIVE)
BioCLIP ViT-B
$\mathbf{f}_{\text{bio}} \in \mathbb{R}^{512}$
Taxonomic Vision
Fast-Exit Routing
896-d Concatenation
$\mathbf{z}_q = [\mathbf{f}_{\text{bio}}, \mathbf{f}_{\text{dino}}]$
Fisher + Text Ensemble
$\mathbf{p} = 0.75\mathbf{p}_{\text{fisher}} + 0.25\mathbf{p}_{\text{text}}$
Trained Fisher Head (W, b) [<10 KB]
Diagnostic Prediction \& ECE
Peak Accuracy: 98.67% | ECE: 0.79%
Temperature Scaling ($T = 0.50$)
Soft-Anchored Transductive Recalibration (3× Anchor)
Online Buffer → Closed-Form $\hat{\mathbf{\Sigma}}^{\text{trans}}$ & $\mathbf{W}^{\text{trans}}$ re-estimation (<22 ms)